

By 2nd Associative Law of Propositions, $(x \in A \wedge x \notin A) \wedge x \in B$.

By Defn. 5, $(x \in A \wedge x \in \bar{A}) \wedge x \in B$.

By Defn. 2, $(x \in A \wedge \bar{A}) \wedge x \in B$.

By Prob-9(b), $x \in \emptyset \wedge x \in B$. By 2nd Commutative Law of Proposition,

$x \in B \wedge x \in \emptyset$. By Defn. 2, $x \in (B \cap \emptyset)$. By Prob 7(b), $x \in \emptyset$.

By Defn. 3 from Section 2.1, $A \cap (B - A) \subseteq \emptyset$.

Let, x be an arbitrary element of $\emptyset$, i.e. $x \in \emptyset$. By defn. of empty set, no such x exists. This can arise when ~~$x \in A \wedge x \notin A$~~ $x \in A \wedge \bar{A}$ by Prob 9(b). By Defn. 2, $x \in A \wedge x \in \bar{A}$. By Defn. 5, $x \in A \wedge x \notin A$.

From Prob 7(b), $x \in \emptyset$ is the same as $x \in B \cap \emptyset$, By Defn. 2, $x \in B \wedge x \in \emptyset$. ~~$\therefore x \in B \wedge (x \in A \wedge x \notin A)$~~ $\therefore x \in B \wedge (x \in A \wedge x \notin A)$. By 2nd Associative Law of Propositions, $(x \in B \wedge x \in A) \wedge x \notin A$. By 2nd Commutative Law of Propositions, $(x \in A \wedge x \in B) \wedge x \notin A$. By 2nd Associative Law of Propositions, $x \in A \wedge (x \in B \wedge x \notin A)$. By Defn. 4, $x \in A \wedge x \in B - A$. By Defn. 2, $x \in A \cap (B - A)$.

By Defn. 3 from Section 2.1, $\emptyset \subseteq A \cap (B - A)$.

$\therefore A \cap (B - A) = \emptyset$ $\square$

e) $A \cup (B - A) = A \cup B$

Solution: Let, x be an arbitrary element of $A \cup (B - A)$, i.e. $x \in A \cup (B - A)$. By Defn. 5, $x \in A \vee (x \in B \wedge x \notin A)$. By Defn. 4, $x \in A \vee (x \in B \wedge \bar{A})$. By Defn. 1, $x \in A \vee x \in (B - A)$. By Defn. 4, $x \in A \vee (x \in B \wedge \bar{A})$. By 1st Distributive Law of Proposition, $(x \in A \vee x \in B) \wedge (x \in A \vee \bar{A})$. By Defn. 1, $(x \in A \cup B) \wedge (x \in A \vee \bar{A})$. By Prob 9(a), $(x \in A \cup B) \wedge (x \in U)$. By Defn. 2, $x \in (A \cup B) \cap U$. By Prob-6(b), $x \in A \cup B$. By Defn. 3 from Section 2.1, $A \cup (B - A) \subseteq A \cup B$.

Let, x be an arbitrary element of $A \cup B$, i.e. $x \in A \cup B$. By Defn. 1, $x \in A \vee x \in B$. By 1st ~~Distributive~~ Law of Propositions, $(x \in A \vee x \in B) \wedge T$. By Defn. of universal set, $(x \in A \vee x \in B) \wedge (x \in U)$. By Prob-9(a), we know $(x \in A \vee x \in B) \wedge (x \in (A \cup \bar{A}))$.